Multifunctional Organic-Inorganic Hybrid Materials for Energy Storage

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Although metal oxides/carbons have attracted huge interest in academic research, they were still relatively limited for their practical applications. Though several refinements have been suggested, the common problem was their synthesis by traditional synthesis ways, which provide little control over design and physical properties of these materials. Their true potential for electrochemical applications is inhabited by our ability to control nano-architectures of these materials.

MOFs on the horizon are poised to take care of these problems. Now, it is possible to design and develop shape, size, porosity, phase and/or conductivity controlled nanoporous metal oxide/carbon by selecting the proper organic and inorganic constitutes and controlling their precursor or templates conditions for the derivation of desired porous metal oxides/carbon materials. These materials possess interesting electrochemical properties in combination with proper electrolyte. The future development of MOF materials is aimed to address challenges in the field to fully unlock their true potential for energy storage applications.